Juan David Muñoz-Bolaños
Rechenauerstraße 42, Innsbruck, Austria (Tirol)
+43 676 9115145
juan.munoz@i-med.ac.at

Silicon Austria Labs (SAL)
Hiring Team
Europastraße 12
9524 Villach, Austria

Subject: Application for Research Engineer – Solid-state quantum systems on chip

Dear Hiring Team,

I am writing to express my strong enthusiasm for the Research Engineer – Solid-state quantum systems on chip position. Concluding my PhD in Optics at the Medical University of Innsbruck and holding an M.Sc. in Photonics, my diverse experimental background in advanced optical systems strongly aligns with your need for innovative hardware and signal testing in applied quantum sensing.

Throughout my PhD and prior work at IPHT Jena, I have taken sole responsibility for designing complex electro-optical systems from the ground up. I routinely integrate high-power continuous-wave/pulsed lasers, optical modulators, and sensitive detection modules (PMTs/Spectrometers) and handle highly photon-starved imaging regimes, mirroring the hardware solutions and experimental precision required for quantum systems-on-chip.

Hardware Integration & Programming:
Modern physics relies on a seamless bridge between optical hardware and software. I develop high-performance Python pipelines and hardware-control scripts to synchronize complex excitation sources with real-time signal processing. This provides me with the prototyping and engineering skills essential to driving novel measurement and control protocols.

Integration & Austrian Work Eligibility:
Currently working as an active researcher in Austria, I am fully integrated into the local culture in Tyrol and hold a certified B2 level in German alongside fluent English. Furthermore, having worked for the last 4 years in Austria as a PhD candidate, I can seamlessly transition my current resident permit to allow full working rights in Austria without requiring standard sponsorship delays.

With my deep passion for experimental physics and an extensive toolkit in optics, spectroscopy, and hands-on laboratory integration, I am highly motivated to tackle the industrial challenges at Silicon Austria Labs. I welcome the opportunity to discuss how my background can directly advance your quantum sensing systems.

Sincerely,

Juan David Muñoz-Bolaños
Optical System Engineer
Medical University of Innsbruck

Juan David Muñoz-Bolaños

Optical System Engineer

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145
juan.munoz@i-med.ac.at | LinkedIn Profile

PROFESSIONAL SUMMARY

Experimental Physicist and Optical Engineer specialized in the architecture, integration, and prototyping of advanced optical and electro-optical platforms. Extensive hands-on laboratory experience integrating high-power lasers, precision metrology, spatial modulators, and highly sensitive detection modules. Proven capability to bridge physics with robust hardware control software (Python) to drive applied research from conceptualization to reliable prototype.

TECHNICAL COMPETENCIES

Applied Physics & Experimental Optics
Substantial expertise in experimental optics, spectroscopy, and custom system design. Demonstrated ability to assemble and optimize complex photon-starved setups, analogous to the precision required in quantum technologies.

Prototyping & Hardware Integration
Hands-on integration of continuous-wave/pulsed lasers, spatial light modulators, acousto-optics, and highly sensitive detection modules (PMTs, Spectrometers). Focus on optimizing signal-to-noise ratios in delicate physical systems.

Software Integration & Control
Advanced Python (NumPy, JAX, Traditional AI), C++, LabVIEW, and MATLAB for hardware interfacing, data acquisition, and real-time signal processing. Experience designing robust software pipelines to synchronize electro-optical hardware with precise measurement protocols.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck — Innsbruck, Austria
PhD Researcher - Optical Engineering
Jan 2022 – Present

- **Microscopy System Architecture:** Designed and constructed advanced Two-Photon microscopy setups from the ground up, directing the full optical integration from laser source to detector.
- **Hardware & Software Control:** Developed rigorous data acquisition and synchronization pipelines using Python, C++, LabVIEW, and MATLAB. Integrated traditional AI for automated image analysis and signal processing of complex electro-optical components.
- **Core Technology Integration:** Integrated and characterized novel core technologies, including adaptive optics hardware, to fundamentally improve spatial resolution in deep, scattering media.
- **Project Management:** Led independent R&D initiatives, taking sole operational responsibility for evaluating new concepts and maintaining complex laboratory prototypes.

Leibniz Institute of Photonic Technology (IPHT) — Jena, Germany
R&D Assistant Researcher
Jun 2020 – Dec 2021

- **Prototype Construction:** Designed and assembled a dispersive 1064 nm fiber probe Raman imaging spectrometer, and calibrated vibrational point-scanning prototypes for clinical tissue applications.
- **Innovation & Evaluation:** Bridged theoretical R&D with practical lab implementation, utilizing traditional AI (supervised tools) for hyperspectral data analysis to streamline experimental characterization.

INTECOL S.A.S. — Medellin, Colombia
Machine Vision System Engineer
Mar 2018 – Mar 2019

- **System Architecture & Component Selection:** Directed the physical design of optical inspection setups by analyzing and selecting appropriate sensors, industrial cameras, illumination schemes, and custom optics.

- **Machine Vision Integration:** Implemented robust, industrial-grade image processing pipelines to ensure precise real-time defect detection and tracking reliability within manufacturing environments.

EDUCATION

Medical University of Innsbruck Austria
Ph.D. in Adaptive Optics & Microscopy *Jan 2022 – Present*

Friedrich Schiller University Jena Germany
M.Sc. in Photonics *2019 – 2021*

University of Cauca Colombia
B.Sc. in Physics Engineering *2012 – 2018*

PUBLICATIONS & AWARDS

- **Awards & Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for the best Colombian professionals.
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer and its application to human bladder resectates. Applied Sciences, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

LANGUAGES

English (Proficient),
German (Intermediate/Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte Medical University of Innsbruck
Head of Institute for Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Dr. Christoph Krafft Leibniz Institute of Photonic Technology
Group leader: Spectroscopy Imaging
christoph.krafft@leibniz-ipht.de

FRIEDRICH-SCHILLER-UNIVERSITÄT JENA
PHYSIKALISCH-ASTRONOMISCHE FAKULTÄT

URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021

Friedrich-Schiller-Universität Jena
2

Prof. Dr. Christian Spielmann
Dekan

Prof. Dr. Silvana Botti
Vorsitzende des Prüfungsausschusses

FRIEDRICH-SCHILLER-
UNIVERSITÄT
JENA

REPUBLICA DE COLOMBIA

PASAPORTE
PASSPORT

Tipo / *Type*	Cod. país / *Country code*	Pasaporte Nº / *Passport No.*
P	COL	AV410884

Apellidos / *Surname*
MUÑOZ BOLAÑOS

Nombres / *Given names*
JUAN DAVID

Nacionalidad / *Nationality*
COLOMBIANA

Fecha de nacimiento / *Date of birth*
04 JUN/JUN 1994

Núm. personal / *Personal No.*
CC1061772278

Sexo / *Sex*
M

Lugar de nacimiento / *Place of birth*
LA CRUZ COL

Fecha de expedición / *Date of issue*
11 DIC/DEC 2018

Autoridad / *Authority*
G. ANTIOQUIA

Fecha de vencimiento / *Date of expiry*
10 DIC/DEC 2028

Firma del titular / *Holder's signature*

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SONSTIGE ANGABEN
GILT BEI MEDIZINISCHE
UNIVERSITÄT INNSBRUCK
AUSSTELLUNGSDATUM
16 05 2024
AUSSTELLENDE BEHÖRDE
BGM. INNSBRUCK
GEBURTSORT
LA CRUZ
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AUT
AUFENTHALTSTITEL
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NACHNAMEN Vornamen
MUÑOZ BOLAÑOS
Juan David
GESCHLECHT
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STAATSANGEHÖRIGKEIT
COL
GEBURTSDATUM
04 06 1994
KARTE GÜLTIG BIS
16 05 2026
ART DES AUFENTHALTSTITELS
NIEDERLASSUNGSBEWILLIGUNG
SONSTIGE ANGABEN
FORSCHER
349461
RESIDENCE PERMIT

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994
in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden

österreich • ösd • INNSBRUCK • schweiz • deutschland

Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA

Europäisches Siegel für innovative Spracheninitiativen 1998

ÖSD ist Mitglied von

ALTE Association of Language Testers in Europe

ösd
österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

CERTIFICATE

Juan Munoz Bolanos

has participated in the

ZEISS European Summer School

Lithography Optics & Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammler

Dr. Thomas Stammler
CTO & Head of Product Strategy

ZEISS

INTECOL
OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos ($1,500,000) plus three hundred thousand pesos ($300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL
NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO
Administrative and Financial Director

Leibniz ipht
LEIBNIZ-INSTITUT für PHOTONISCHE TECHNOLOGIEN

AGREEMENT FOR GUESTS

to carry out a practical training

with

Mr.	**Juan David Munoz Bolanos**	**Date of birth: 1994-06-04**
temporarily living at	**Emil-Wölk-Str. 7, 07747 Jena**	**(guest),**

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2021-01-01 up to 2021-09-30 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolanos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 13 November 2020

F. Sondermann

Prof. J. Popp	F. Sondermann	Juan David Munoz Bolanos
Chairman	Executive Member	Guest
of Leibniz-IPHT	of Leibniz-IPHT	

Mitglied der Leibniz-Gemeinschaft

Leibniz | ipht
LEIBNIZ-INSTITUT für PHOTONISCHE TECHNOLOGIEN

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Jena, 13 November 2020

Prof. J. Popp Chairman of Leibniz-IPHT	F. Sondermann Executive Member of Leibniz-IPHT	Juan David Munoz Bolanos Guest